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$y = \sqrt[3]{\frac{3x^4}{4} + C}$

2) $\tan y = x + C$

$y = \tan^{-1} (x + C)$

4) $\frac{e^{2y}}{2} = x^2 + C_1$

$y = \frac{\ln(2x^2 + C)}{2}$

5) $\frac{y^3}{3} = x^2 + \frac{1}{3}$

$y = \sqrt[3]{3x^2 + 1}$

6) $e^y = 2e^x + 3$

$y = \ln (2e^x + 3)$

8) $\frac{y^3}{3} = e^x + \frac{1}{3}$

$y = \sqrt[3]{3}e^x + 1$

9) $\cos y = x - 3$

$y = \cos^{-1}(x - 3), 2 < x < 4$

$\frac{e^{2y}}{2} = x^2 - \frac{3}{2}$

$y = \frac{\ln(2x^2 - 3)}{2}, x > \frac{\sqrt{6}}{2}$

11) $\frac{e^{2y}}{2} = x^3 - \frac{1}{2}$

$y = \frac{\ln(2x^3 - 1)}{2}, x > \frac{\sqrt[3]{4}}{2}$

12) $\frac{y^3}{3} = x + \frac{x^3}{3}$

$y = \sqrt[3]{x^3 + 3x}, x < 0$

Find the general solution of each differential equation.

2) $\frac{dy}{dx} = \frac{1}{\sec^2 y}$

3) $\frac{dy}{dx} = 3e^{x-y}$

5) $\frac{dy}{dx} = \frac{2x}{y^2}$, $y(2) = \sqrt[3]{13}$

6) $\frac{dy}{dx} = 2e^{x-y}$, $y(-3) = \ln \frac{3e^3 + 2}{e^3}$

7) $\frac{dy}{dx} = \frac{1}{\sec^2 y}$, $y(3) = 0$

8) $\frac{dy}{dx} = \frac{e^x}{y^2}$, $y(-1) = \frac{\sqrt[3]{e^3 + 3e^2}}{e}$

9) $\frac{dy}{dx} = -\frac{1}{\sin y}$, $y(3) = \frac{\pi}{2}$

11) $\frac{dy}{dx} = \frac{3x^2}{e^{2y}}$, $y(1) = 0$

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